GEETHANJALI INSTITUTE OF SCIENCE & TECHNOLOGY

(AUTONOMOUS)

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QUESTION BANK (DESCRIPTIVE)	
Subject Name with Code	**SUSTAINABILITY IN ENGINEERING PRACTICES (23A0151T)**
Course & Branch	**B. Tech & CSE, EEE, AIML, CS, DS**
Year& Semester	**III & II**
Regulation	**RG23**

UNIT - I

S. No.	Question	[BT Level] [CO] [Marks]
2 Marks Questions (Short)		
1.	Define sustainability.	**L1 CO1 2M**
2.	List the pillars of sustainability.	**L1 CO1 2M**
3.	Define Economic sustainability.	**L1 CO1 2M**
4.	Define carbon cycle.	**L1 CO1 2M**
5.	List the processes of fast carbon cycle.	**L1 CO1 2M**
6.	List the major construction materials contributing to CO_2 emissions.	**L1 CO1 2M**
7.	Define concrete and steel.	**L1 CO1 2M**
Descriptive Questions (Long)		
8.	Explain the concept of sustainability and its importance in civil engineering.	**L2 CO1 10M**
9.	Describe the carbon cycle with a neat diagram.	**L2 CO1 10M**
10.	Discuss the role of construction Materials in detail .	**L2 CO1 10M**
11.	Explain CO_2 emissions from cement and other materials.	**L2 CO1 10M**
12.	Explain about concrete and steel.	**L2 CO1 10M**
13.	Write the difference between concrete and steel.	**L2 CO1 10M**

UNIT - II

S. No.	Question	[BT Level] [CO][Marks]
2 Marks Questions (Short)		
1.	What are low-VOC materials.	**L1 CO2 2M**
2.	Define indoor air quality.	**L1 CO2 2M**
3.	What is No/Low cement concrete.	**L1 CO2 2M**
4.	Define recycled aggregate.	**L1 CO2 2M**
5.	What is durability and why is it important.	**L1 CO2 2M**
6.	What are manufactured aggregates.	**L1 CO2 2M**
7.	What is Life Cycle Assessment.	**L1 CO2 2M**
8.	Define embodied energy.	**L1 CO2 2M**
Descriptive Questions (Long)		
9.	Explain briefly about construction materials in maintaining good indoor air quality.	**L2 CO2 10M**
10.	Describe No/Low Cement Concrete and its sustainability benefits.	**L2 CO2 10M**
11.	Explain recycled aggregates & their properties.	**L2 CO2 10M**
12.	Discuss the need for quality control in sustainable materials.	**L2 CO2 10M**
13.	Explain the concept of durability and its role in sustainability.	**L2 CO2 10M**
14.	What is LCA? Explain stages of life cycle assessment.	**L2 CO2 10M**

UNIT - III

S. No.	Question	[BT Level] [CO][Marks]
2 Marks Questions (Short)		
1.	Define embodied energy & give examples of embodied energy.	**L1 CO3 2M**
2.	What is primary energy? give examples for primary energy.	**L1 CO3 2M**
3.	Define operational energy.	**L1 CO3 2M**
4.	What is life cycle energy use.	**L1 CO3 2M**
5.	List the components of embodied energy.	**L1 CO3 2M**
6.	Define energy payback period.	**L1 CO3 2M**
Descriptive Questions (Long)		
7.	Explain briefly about the components of Embodied energy.	**L2 CO3 10M**
8.	Explain about primary energy.	**L2 CO3 10M**
9.	Discuss the relationship between embodied energy and operational energy.	**L2 CO3 10M**
10.	Explain about Life Cycle Energy use and also how to calculate LCE step by step procedure.	**L2 CO3 10M**
11.	A conventional building having EE of 3000MJ/m^2 and OE of 150 Kwh/m^2 /year and high-performance building having EE of 4500MJ/m^2; OE of 50 Kwh/m^2/year calculate total life cycle every 50 Years and also Break-even time for extra EE	**L3 CO3 10M**
12.	Calculate Embodied energy of a non-condition building with construction materials include Reinforced concrete (Cement=320Kg, Fine aggregate=660Kg, Coarse Aggregate=1200Kg and Steel=50Kg) embodied coefficients of each material Cement=4.5 MJ/Kg, Fine aggregate and coarse aggregate=0.5MJ/Kg, Steel=30MJ/Kg.	**L3 CO3 10M**

UNIT - IV

S.No.	Question	[BT Level] [CO][Marks]
2 Marks Questions (Short)		
1.	What is ECBC.	**L1 CO4 2M**
2.	Define OTTV.	**L1 CO4 2M**
3.	What is a green building.	**L1 CO4 2M**
4.	Mention features of LEED rating system.	**L1 CO4 2M**
5.	Define TERI–GRIHA.	**L1 CO4 2M**
6.	What is a Zero Energy Building.	**L1 CO4 2M**
Descriptive Questions (Long)		
7.	Explain various strategies to control energy use in buildings.	**L2 CO4 10M**
8.	Explain briefly about ECBC code and its mandatory requirements.	**L2 CO4 10M**
9.	Explain about OTTV concepts.	**L2 CO4 10M**
10.	Discuss about LEED and TERI in detail.	**L2 CO4 10M**
11.	Explain about GRIHA.	**L2 CO4 10M**
12.	Explain briefly about Role of insulation and thermal properties of construction Materials.	**L2 CO4 10M**
13.	Describe the concept of Zero Energy Building with examples.	**L2 CO4 10M**
14.	Discuss about Influence of Moisture content and modeling.	**L2 CO4 10M**

15.	Calculate OTTV for a following building having of Wall area $A_w = 80\ m^2$; Glass area $A_f = 20\ m^2$; Total area $A_t = 100\ m^2$; Wall U-value $U_w = 2.0$; Glass U-value $U_f = 5.8$; $\Delta T = 10°C$; Solar Factor $SF = 250\ W/m^2$; SHGC = 0.60. explain whether building meets ECBC compliance	**L3 CO4 10M**

UNIT - V

S.No.	Question	[BT Level] [CO][Marks]
2 Marks Questions (Short)		
1.	Define non-renewable energy sources.	**L1 CO5 2M**
2.	List any two environmental impacts of coal mining.	**L1 CO5 2M**
3.	Define greenhouse gases.	**L1 CO5 2M**
4.	What is global warming.	**L1 CO5 2M**
5.	What is acid rain.	**L1 CO5 2M**
6.	Define nuclear energy.	**L1 CO5 2M**
7.	Mention two regional impacts of temperature change.	**L1 CO5 2M**
Descriptive Questions (Long)		
8.	Explain major non-renewable energy sources and their environmental impact.	**L2 CO5 10M**
9.	Describe the greenhouse effect	**L2 CO5 10M**
10.	Explain the causes and effects of global warming.	**L2 CO5 10M**
11.	Discuss the causes, effects, and control methods of acid rain.	**L2 CO5 10M**
12.	Explain nuclear energy, advantages, and Environmental safety concerns.	**L2 CO5 10M**
13.	Describe regional impacts of temperature change.	**L2 CO5 10M**
14.	Write about global temperature rise and climatic patterns.	**L1 CO5 10M**

Signature of the Staff:

Signature of Department Academic Committee Member 1:

Signature of Department Academic Committee Member 2:

Signature of Department Academic Committee Member 3: